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C.3 Order Statistics

L1 Order Statistics

- Suppose $X_1, \dots X_n$ are i.i.d samples. Let $Y_1 = \min\{X_1, \dots X_n\}$ and let $Y_n = \max\{X_1, \dots X_n\}$.

then,

$$P(\max\{X_1, \dots X_n\} \leq x) = P(Y_n \leq x) = [F_X(x)]^n$$

$$P(\min\{X_1, \dots X_n\} > x) = P(Y_1 \geq x) = [P(X > x)]^n$$

- If $X_1 \dots X_n$ are iid discrete RV, then

$$P(Y_n = x) = P(Y_n \leq x) - P(Y_n \leq x-1)$$
$$= [F_X(x)]^n - [F_X(x-1)]^n$$

$$P(Y_1 = x) = P(Y_1 \geq x) - P(Y_1 \geq x+1)$$
$$= (P[X \geq x])^n - [P(X \geq x+1)]^n$$

L2 General Order Stats

- Suppose if we have n data points, denoted as $X_1 \dots X_n$, where X_i are iid random variables. We can sort the n data points from smallest to largest

$$X_{(1)} < X_{(2)} < X_{(3)} < X_{(4)} < \dots < X_{(n)}$$

$$Y_1 < Y_2 < Y_3 \text{ --- } < Y_n$$

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The i^{th} smallest data point is c/d the i^{th} order statistic and is denoted either as $X_{(i)}$ or Y_i.

- So, Y_1 is the minimum

$$P[Y_1 < y] = P(\text{at least one } X_i \text{ is } \leq y)$$
$$= 1 - P(\text{all } X_i > y)$$
$$= 1 - (1 - P(X_i < y))^n$$
$$= 1 - (1 - F_X(y))^n$$

Y_n is the max value

$$P(Y_n \leq y) = P(\text{all } n \text{ points are } \leq y)$$
$$= [F_X(y)]^n$$

- The density of a general order stat

$$f_i(y) = \text{density of } Y_i$$

$$f_i(y)\,dy = P[y \leq Y_i \leq y + dy]$$